

DATS 598: Data Science Capstone

Project Proposal

A Statistical Truth Detection System

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Contents

Exploratory Data Analysis	2
Model Tasks Exploration	3
Baseline Models	3

Exploratory Data Analysis

Data Understanding

In this project the data utilized will essentially be from two sources. The first source is wikipedia dataset that is pulled from a recent wikipedia dump. This contains the entirety of wikipedia articles that are then embedded using a selected model and stored in a vector database locally. The second source of data is collected from LLM outputs. This data is parsed via a text processes pipeline to extract the claims made. These two sources serve as the backbone for the truth detection system.

The potential issues certainly lie within the NLP, text cleaning processes as well as the variability coming out of the LLM. Assuming the text processing is done well, alignment with evidence retrieved should enable a straightforward verification of claims found, however, if either the claims found or evidence retrieved are poorly done, the system will struggle to perform well.

Data Cleaning and Preprocessing

The text preprocessing pipeline is designed to do a few things: clean text, detect claims, and extract claims from said text. This is an object oriented process so if multiple text blocks were to run at a time, each would be handled by its own instance and enable parallel processing with text-specific evidence associated with it.

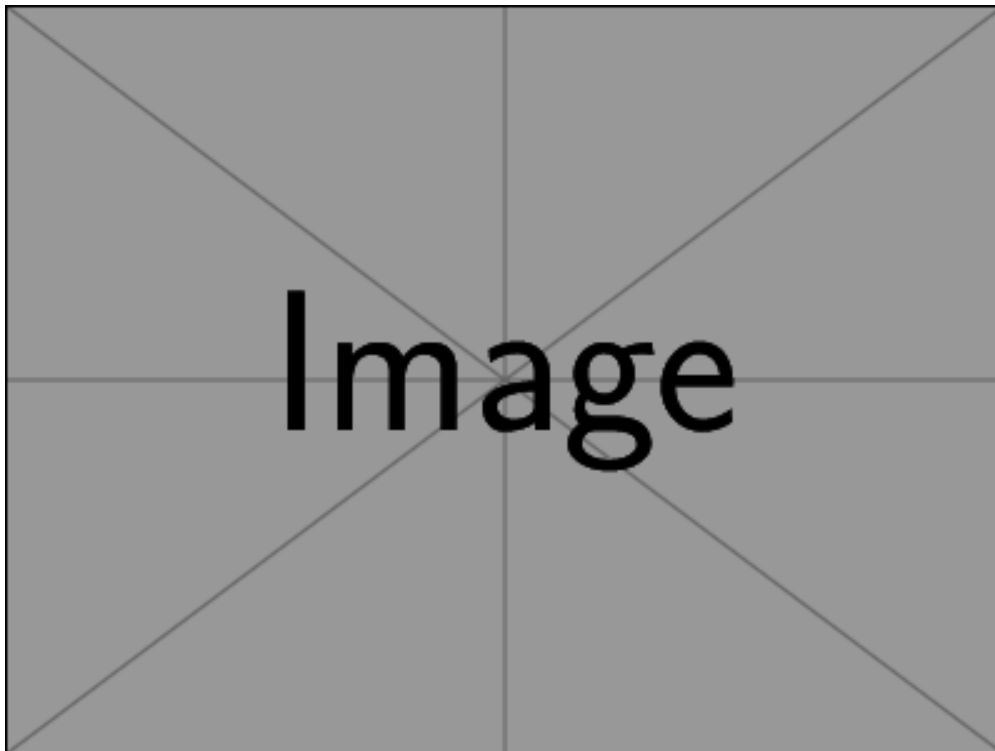


Figure 1: Cleaning Process.

As shown in Figure 1, is an example line.

Exploratory Data Analysis

Perform descriptive statistics and visualizations to understand the data characteristics. Analyze the distribution of features and target variables. Identify any patterns, correlations, or potential relationships within the data. Include relevant plots, charts, and visualizations to support your findings.

Model Tasks Exploration

Task Definition

Clearly define the problem or task you are trying to solve (e.g., classification, regression, clustering). Explain the rationale behind choosing this specific task.

Feature Engineering

Describe the process of selecting and transforming relevant features for your task. Discuss any domain-specific knowledge or techniques applied to feature engineering.

Model Selection

Explore different model architectures or algorithms suitable for your task. Justify the choice of models based on the data characteristics and task requirements.

Baseline Models

Model Implementation

Provide details on the implementation of your baseline models. Include code snippets or references to the code used for training and evaluation.

Model Evaluation

Define the evaluation metrics used to assess model performance. Report the performance of your baseline models on the training, validation, and test sets.